



DiffShield: A Tri-Modal Adversarial Framework for Robust Immunization Against Deepfake Generation

23CSE399 – Project Phase 1 | Panel Review 2 | Team 15

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2. Literature Survey

- **Survey Scope:** Adversarial ML & Facial Privacy against Generative AI (2018–2025).
- **Critical Analysis of SOTA:**

Paper	Method	Key Findings	
		Strength	Critical Gap
PhotoGuard [1]	VAE Encoder Attack	Disrupts image generation	Fails under JPEG compression
AdvI2I [2]	CLIP Semantic Attack	High semantic disruption	Offensive tool; fails vs. ControlNet
Anti-DreamBooth [4]	Fine tuning Disruption	Prevents model personalization	Fails against zero shot ControlNet
DiffAM [3]	Adversarial Makeup	Protects facial privacy	Visibly alters face appearance
TUAP [10]	Universal CLIP Perturbation	77.4% ASR across CLIP models	Mitigated by test time defenses [7]

- **Survey Synthesis:** Every reviewed defense attacks one perception pathway. None simultaneously addresses (a) compression robustness and (b) geometric override via ControlNet. This dual gap is the precise target of DiffShield.

3. Originality of Work

Novelty: Tri-Modal Adversarial Optimization (DiffShield)

We extend foundational single-modality attacks into a **unified joint optimization**, integrating:

- ➊ **Visual Collapse (VAE):** Maximizes MSE in the latent space, destroying the spatial structure before the UNet processes it.
- ➋ **Semantic Erasure (CLIP):** Moves image embeddings toward an unrelated concept, severing the identity language link.
- ➌ **Structural Sabotage (Canny):** Injects fake micro-edges to break the clean facial stencil, disrupting ControlNet geometric constraints.

Key Differentiator 1 - EoT Loop:

Unlike PhotoGuard [1], we embed differentiable JPEG/crop transformations (via Kornia) *inside* the gradient loop, so the learned noise mathematically survives social media compression.

Key Differentiator 2 - XAI Validation:

We apply **Grad-CAM** on cross-attention layers to produce heatmaps that visually prove the AI's spatial focus has been scattered away from facial features.

4. Hypothesis, Challenges & Deliverables

Hypothesis (CO1)

Jointly attacking the VAE, CLIP, and Canny pathways under EoT constraints will produce adversarial noise that resists ControlNet geometric overrides in cases where all single-modality defenses fail.

Key Technical Challenges:

- Balancing three competing loss gradients to prevent one modality from dominating.
- Ensuring adversarial noise survives JPEG compression without explicit post processing.
- Scaling PGD optimization to full resolution CelebA-HQ images within feasible GPU compute.

Project Deliverables:

- 1 **DiffShield Pipeline:** PyTorch module producing imperceptible immunized images (verified PSNR > 40 dB, SSIM > 0.95).
- 2 **Robustness Benchmark:** Quantitative ASR measurements showing DiffShield outperforms PhotoGuard and Anti-DreamBooth baselines under SD 1.5 + ControlNet attack conditions.
- 3 **XAI Evidence Portfolio:** Grad-CAM cross attention heatmaps visually documenting the spatial scattering of the generative model's attention away from coherent facial features.

5. Study Methodology & Execution Plan

Modular research-to-implementation strategy across four subdomains:

Subdomain 1 - Data & EoT Pipeline

- CelebA-HQ dataset ingestion and preprocessing.
- Differentiable JPEG/crop augmentations via `Kornia` to implement the EoT layer inside the forward pass.

Subdomain 2 - Optimization Engine

- PyTorch autograd: freeze model weights, compute input-level gradients.
- Custom PGD loop with L_∞ clamping ($\epsilon = 8/255$).

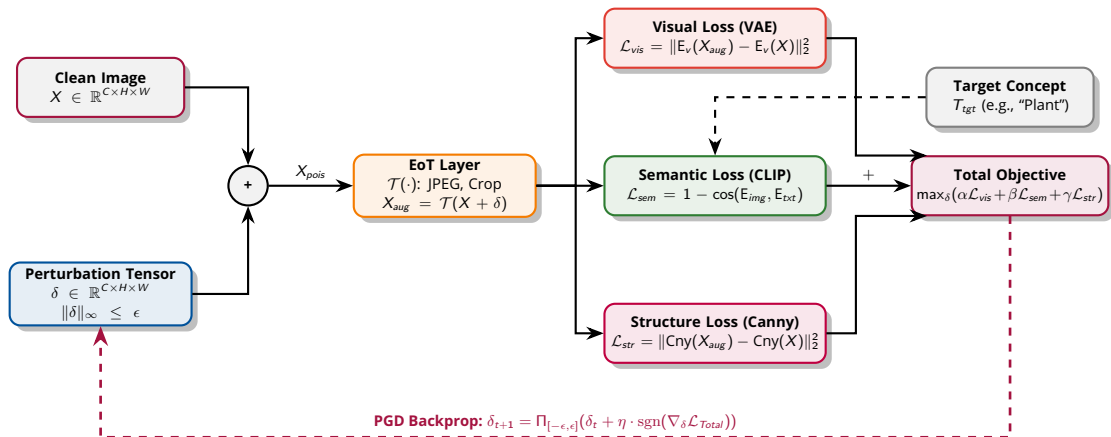
Subdomain 3 - Multi-Modal Integration

- HuggingFace `diffusers`: load frozen AutoencoderKL, CLIP vision encoder, differentiable Canny detector.
- Compute $\mathcal{L}_{vis}$, $\mathcal{L}_{sem}$, $\mathcal{L}_{str}$ simultaneously.

Subdomain 4 - Evaluation & XAI

- Automated metric scripts: PSNR, SSIM, LPIPS, ASR (via ArcFace).
- Cross-attention gradient extraction for Grad-CAM heatmaps.

6. Technical Architecture — DiffShield Pipeline



7. Algorithmic Justification

Algorithm Selection:

- **Multi-Objective PGD:** Model weights of SD and CLIP are frozen. Only the input noise tensor δ is updated, this is input-space optimization, not model fine-tuning.
- **EoT Layer:** A new random transformation $t \sim \mathcal{T}$ is sampled at every forward pass, so gradients learn to be robust across a distribution of compressions, not a single fixed image.
- **Why PGD over FGSM?** PGD's iterative multi-step update produces tighter, more effective perturbations within the L_∞ bound than single-step FGSM.

Design Constraints & Justifications:

- **Tri-Modal Joint Loss:** Eliminates single points of failure. An attacker bypassing the VAE pathway is still blocked by CLIP and Canny simultaneously.
- **L_∞ Clamping ($\epsilon = 8/255$):** Industry standard imperceptibility bound. Strictly enforced after every PGD step via $\Pi_{[-\epsilon, \epsilon]}$.
- **Weighted Objective (α, β, γ):** Learnable scalar weights prevent gradient dominance from any single modality.

8. Phase 1 Findings & Completed Works

Research Findings:

- **Gap Confirmed:** Demonstrated that purely latent defenses [1] are bypassed by ControlNet spatial conditioning, and that single modality semantic attacks [10] are mitigated by test-time defenses [7].
- **Dual Vulnerability Mapped:** Identified that a robust defense must simultaneously address the compression axis *and* the geometric override axis two orthogonal failure modes no prior work jointly tackles.

Design Completions:

- **Mathematical Formulation:** Derived the Tri-Modal joint objective ($\mathcal{L}_{vis}$, $\mathcal{L}_{sem}$, $\mathcal{L}_{str}$) with weighted scalar coefficients.
- **Constraint Specification:** Finalized PGD update rule, EoT sampling strategy via Kornia, and L_∞ boundary ($\epsilon = 8/255$).
- **Architecture Blueprint:** Complete pipeline design across PyTorch, HuggingFace Diffusers, and CelebA HQ dataset ready for Phase 2 implementation.

9. Implementation Timeline

Semester 7 (June–Dec 2026): Phase 2 - Development

- Configure GPU environment; preprocess CelebA-HQ dataset.
- Implement individual loss modules (VAE, CLIP, Canny).
- Build PGD optimization loop with L_∞ projection.
- Integrate EoT layer via Kornia.
- Hyperparameter tuning for α, β, γ weights.

Semester 8 (Dec–Apr 2027): Phase 3 - Validation

- Automated benchmarking vs. SD 1.5 + ControlNet.
- Compute PSNR, SSIM, LPIPS (stealth) and ASR, CLIP-Score (robustness).
- Generate Grad-CAM cross-attention heatmaps.
- Baseline comparison vs. PhotoGuard and Anti-DreamBooth.
- Thesis compilation; target research publication.

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Thank You